

Integration by Parts

ANSWER KEY



The integration by parts formula

- 1 Evaluate $\int x e^x dx$ using integration by parts.
Let $u = x$, $dv = e^x dx$: $= x e^x - \int e^x dx = x e^x - e^x + C$.
- 2 Evaluate $\int x \cos x dx$ using integration by parts.
Let $u = x$, $dv = \cos x dx$: $= x \sin x - \int \sin x dx = x \sin x + \cos x + C$.
- 3 Evaluate $\int \ln x dx$ using integration by parts (with $dv = dx$).
Let $u = \ln x$, $dv = dx$: $= x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - x + C$.

Repeated and definite applications

- 4 Evaluate $\int x^2 e^x dx$, applying integration by parts twice.
First pass ($u = x^2$): $x^2 e^x - 2 \int x e^x dx$. Second pass gives $\int x e^x dx = x e^x - e^x$. Total:
 $x^2 e^x - 2 x e^x + 2 e^x + C$.
- 5 Evaluate $\int_0^1 x e^{2x} dx$ exactly.
Let $u = x$, $dv = e^{2x} dx$: $\left[\frac{x}{2} e^{2x} - \frac{1}{4} e^{2x} \right]_0^1 = \left(\frac{e^2}{2} - \frac{e^2}{4} \right) - \left(0 - \frac{1}{4} \right) = \frac{e^2}{4} + \frac{1}{4}$.